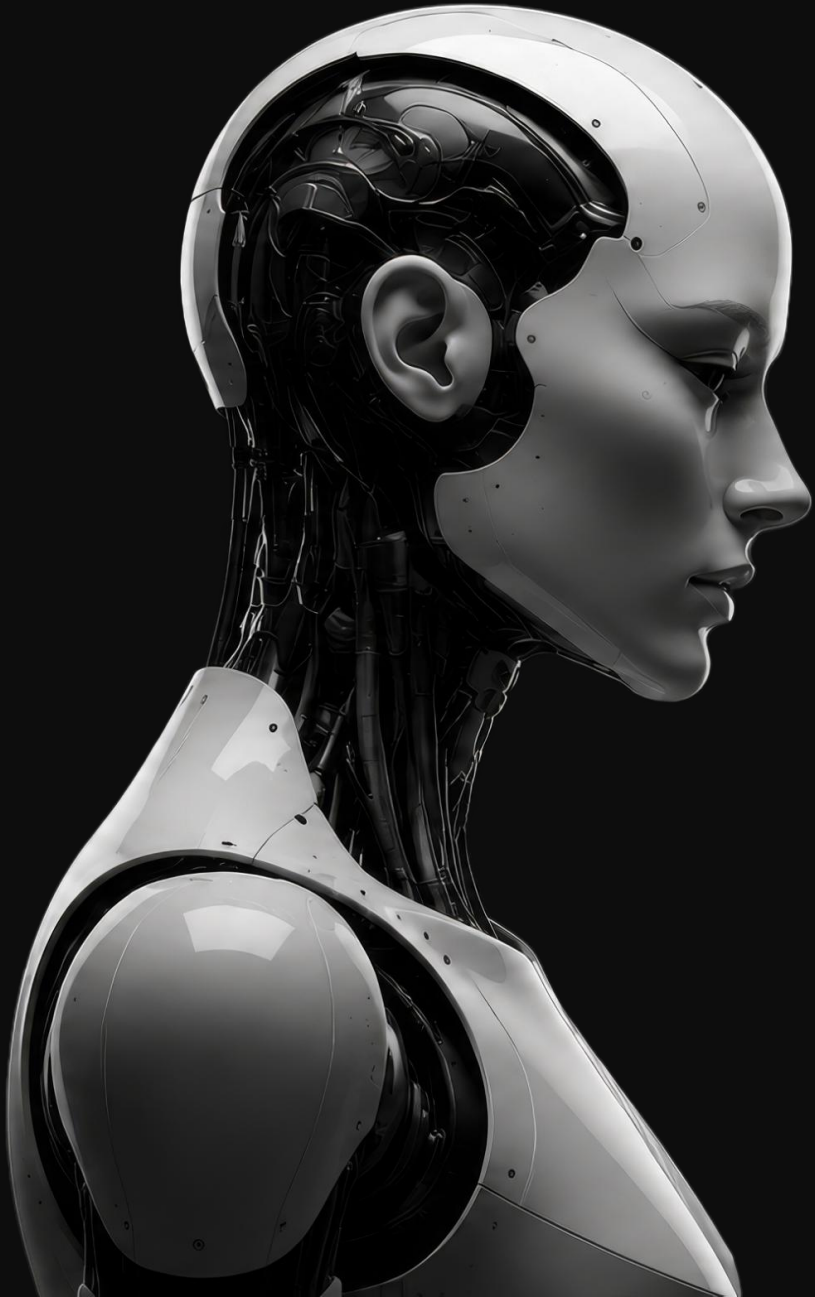




Scaling Physical AI through Robot Foundation Models

2026 Investor Relations

2000

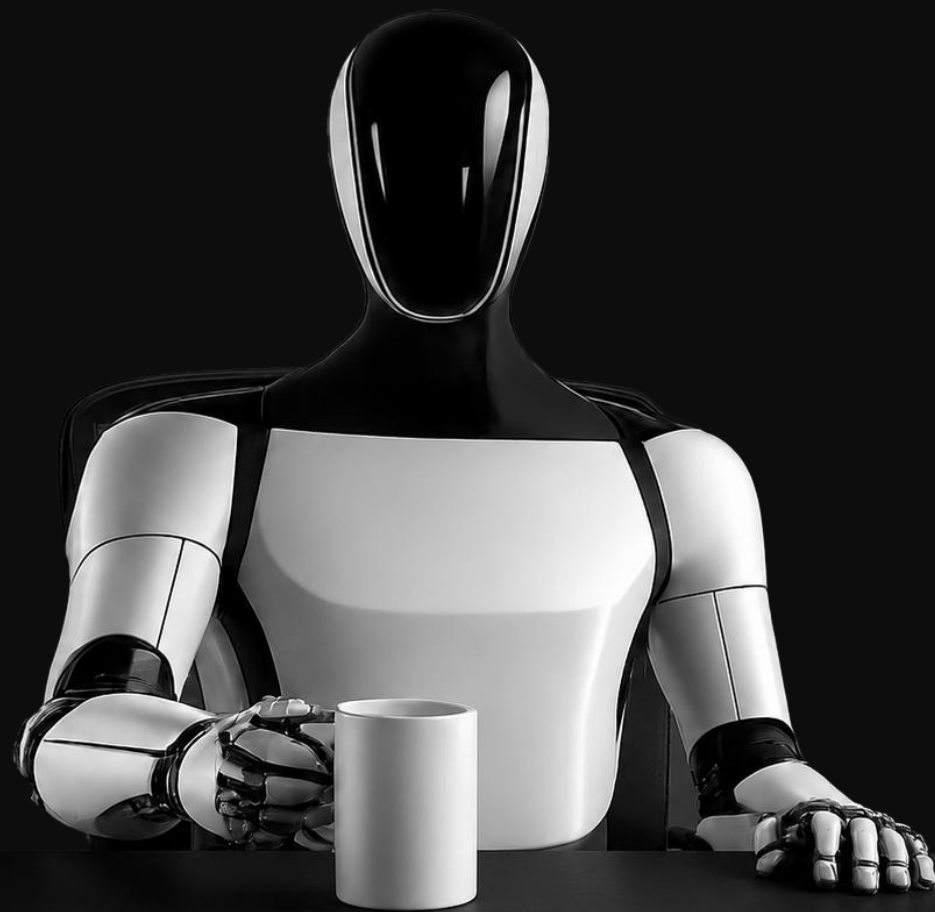
47.6

1



sequor
robotics

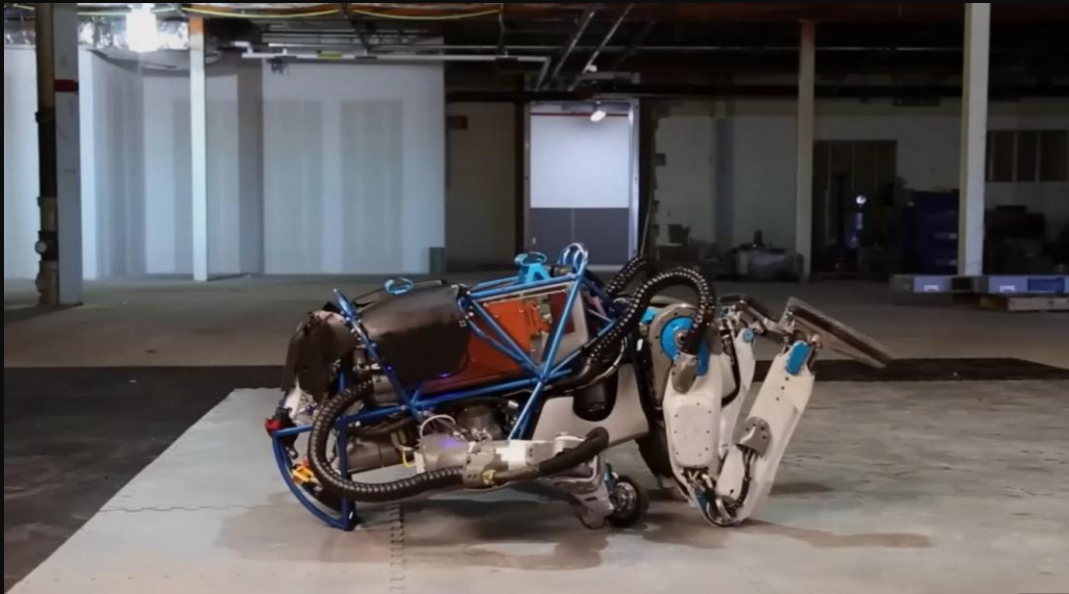
2000 Data
47.6 Hours
1 Task



Training robots needs vast amounts of data and vast amounts of time.

**Ten years have passed, but what robots can actually do
in the real world hasn't changed much.**

2016 Boston Dynamics Atlas Demo Video

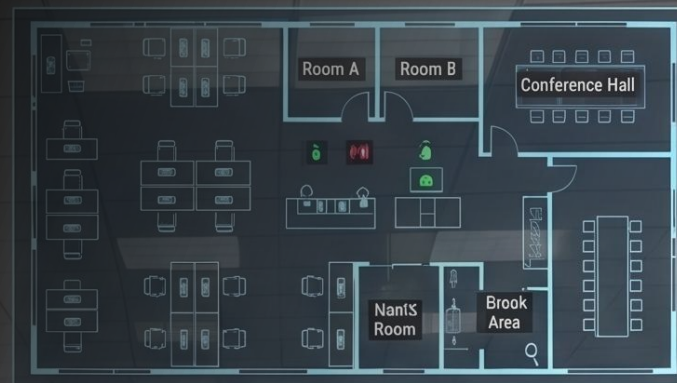


2025 Boston Dynamics Atlas Demo Video



Sequor Robotics, where data becomes robot intelligence

Beyond the limits of 2D,
we build and research 3D data
so robots can understand the world humans actually
live in — shifting the paradigm of robot intelligence.



 OFFICE_TEMP: 22.5°C

 CO2_LEVEL: 600 ppm

 OCCUPANCY: 18/25

 WI-FI_SIGNAL: EXCELLENT


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Just as LLMs grew by training on massive amounts of data, robots too can only advance by learning from massive amounts of data.

2015



Rule-Based Systems

Operating through combinations of human-defined rules and modules

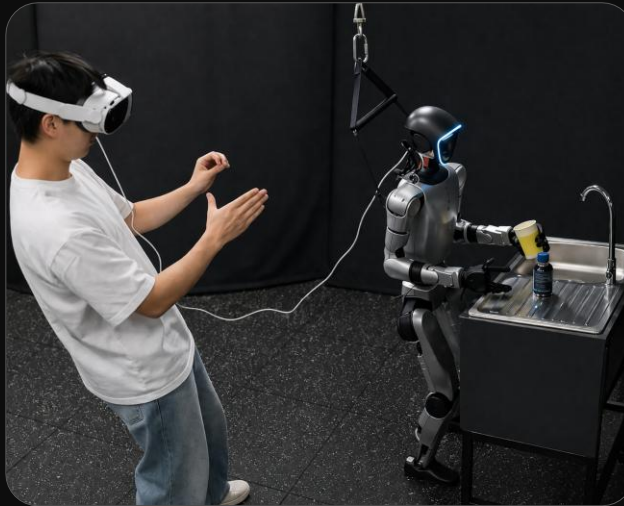


2021



Imitation Learning from Real Data

Collecting real-world data and training via teleoperation



2024



2025



The Limits of Real Data

Despite growing data volumes, generalization continues to fall short

Emerging approaches to robot data collection

Egocentric Video



Collecting manipulation data from the robot's perspective using a first-person camera mounted on the robot arm.

Scene Generation



Collecting large-scale, diverse datasets by automatically generating various environments and object placements.

Human motion capture

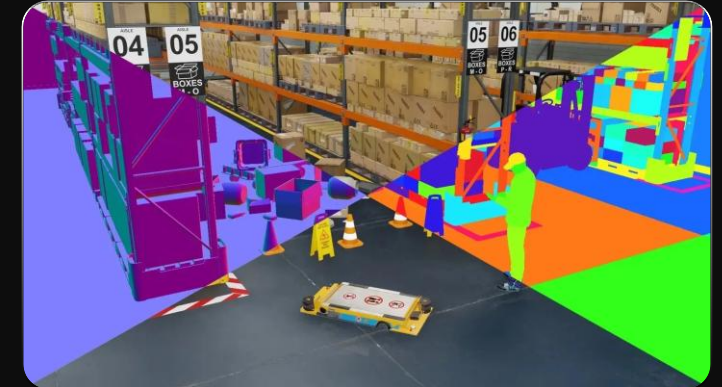


Capturing human motions directly to teach robots natural movement patterns.



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Simulation



Rapidly generating large-scale training data in virtual environments without physical constraints.

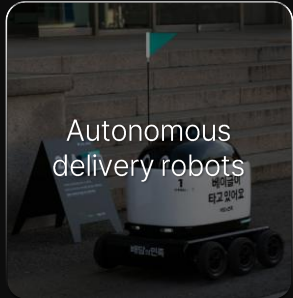
Why Simulation?

Overwhelming Data Volume & Cost Efficiency

Hundreds to thousands of scenarios running in parallel, 24/7				Minimal human involvement, high margins			
Method	Data Volume	Usability	Key Costs				
Simulation	Unlimited	Moderate (sim-to-real gap)	- GPU/compute infrastructure - Simulation environment build (one-time upfront cost) - Software licensing				
Motion Capture	Low	Easy	- MoCap system setup: multiple cameras, calibrated studio space - Data processing engineer labor				
Egocentric Video	High	Hard	- Collection devices: head-mounted cameras, many participants - Labeling costs				
Scene Generation	High	Hard (physics inaccuracy)	- GPU compute: training generative models and running large-scale inference - Generative model licensing / API fees				
Teleoperation	Moderate	Easy	- Labor: skilled operators required for every hour of operation - Robot hardware - Teleoperation interface equipment				

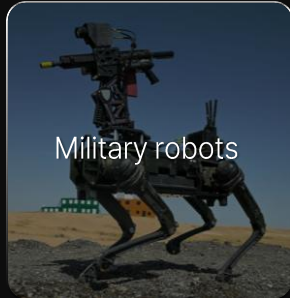
Every robotics company needs Simulation data

Safety Critical — High Risk Scenarios



Autonomous delivery robots

on-road collision



Military robots

real combat environments can't be recreated



Disaster response & firefighting robots

fire and structural collapse sites

Human Interaction — Human Variables



Healthcare & rehabilitation robots

Diverse physical conditions



Caregiving & companion robots

unpredictable behavior patterns



Collaborative robots (cobots)

shared spaces with humans, safety required



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Industrial — Large Scale Data Generation



Manufacturing robots

pick & place, assembly, hundreds of thousands of repetitions



Logistics & warehouse automation

unlimited SKU types, many environmental variables



Agricultural robots

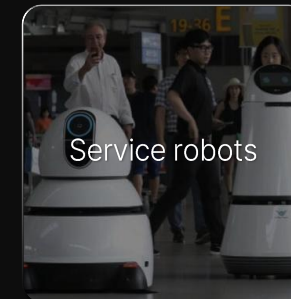
diverse crops and weather conditions

General Intelligence — Complex Environments



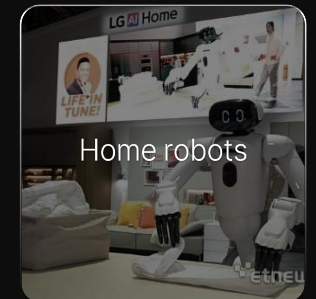
Humanoids

generalized tasks



Service robots

hotels, offices



Home robots

different layouts in every home, multi-step tasks

Customer Pain Point

Even when simulation is needed,

QA/QC Outsourcing Costs

Real robot damage during testing

Testing & debugging 60% of the job

Hardware delays
Compress development timelines

adoption is expensive and painful.

Source: Interviews with engineers at 10 robotics companies (March 2026)

High Cost
Outsourced digital twin \$1M+

- 3D environment build outsourcing costs
- 3D asset purchasing costs

Wasted Time
Outsourcing communication 2hrs/day

- Scene setup and variation work takes days

Poor Training Performance
Useless a 'Pretty 3D map'

- Doesn't reflect real conditions (weather, surfaces)
- Hard to replicate factory/field-level sim fidelity
- Sim gains don't carry over to reality

High Barrier to Entry
Isaac Sim — powerful, but nobody actually uses it

- Most developers have never touched it
- High learning curve (training, time, expertise)
- No accessible UI for non-experts

Solution

Sequor Robotics SW product "Real 3D Gym" — lowering the barrier to simulation adoption

Key Features 1

Scan customer's space with a camera to create the learning-friendly environment

Key Features 2

Custom robot / sensor plugin

Key Features 3

Full end-to-end features for robot learning (tasks, training methods, AI models)



Real 3D Gym Workflow

Simplifying a complex, manual-heavy workflow

STEP 01
Real

Collect real-world spatial data



Scan the physical space using cameras

STEP 02
Real

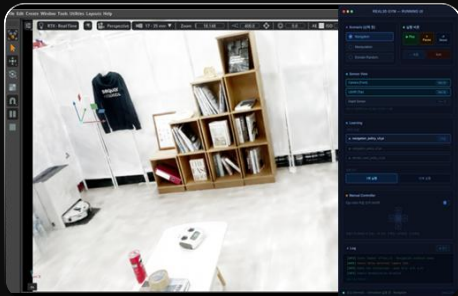
Convert real space into a 3D virtual environment



Generate a 3D space ready for robot learning

STEP 03
Simulation

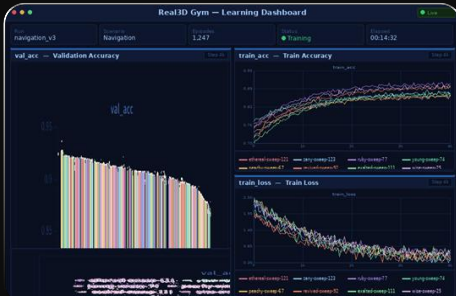
Set up the Real3D Gym environment



Set up learning-ready environment directly in the simulator

STEP 04
Simulation

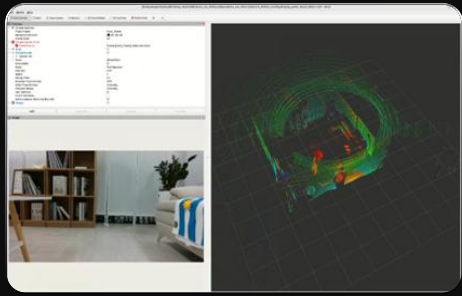
Iterative training in a virtual environment



Optimize robot behavior through thousands of simulations

STEP 05
Real

Deploy learned policies to real robots



Transfer virtual training results to the actual robot

Future Use Case

Sequor's software

— expanding what humanoids can do



Home — Cooking Assistant

Assisting with meal preparation and cleanup in kitchens and dining areas.



1X NEO

home humanoid performing household chores



Scenario Flow



Task

Total ~160 Tasks

Navigation 20 TASK Moving to the fridge and dining table; obstacle avoidance	Pick & Place 80 TASK Picking up ingredients, arranging utensils, clearing and organizing dishes	Open 30 TASK Opening the fridge, cabinets, drawers, and trash cans	Close 30 TASK Closing the fridge, cabinets, drawers, and trash cans
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Deliverables for Robotics Clients

World Files 100 Environments Home environment scans	3D Assets: Furniture Various sizes Kitchen furniture assets	3D Assets: Food 150+ Items Ingredients and cooking tools
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Future Use Case

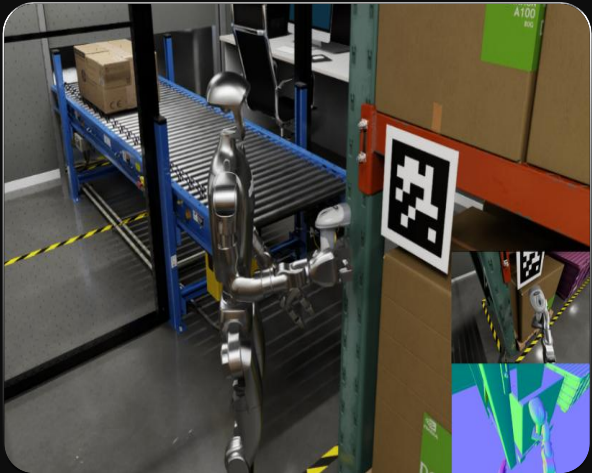
Sequor's software — expanding what humanoids can do

Apptроник Apollo
built for logistics & manufacturing

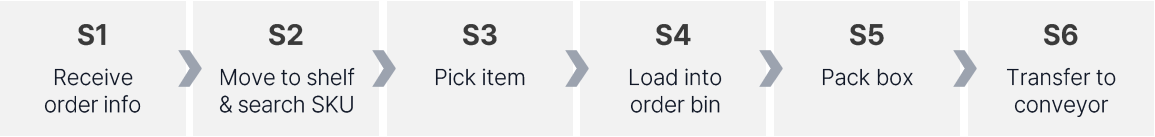


Logistics DAS — SKU Picking & Packing by Order

Shelf picking → Order bin loading → Box packing → Conveyor transfer



Scenario Flow



Task

Total ~200 Tasks				
Navigation 15 TASK Move between shelves and workstations	Pick & Place 100 TASK SKU retrieval & precise placement	Sorting 40 TASK Order bin sorting, mislabel prevention	Packing & Transfer 25 TASK Box packing, conveyor transfer	Cleanup 20 TASK Box assembly, taping, returns

Deliverables for Robotics Clients

World File 1 environment DAS warehouse scan	Structures 15 items Shelves, bins, conveyors	SKU Products 100 SKUs Product packages	Packaging 20 items Boxes, raw materials
---	--	--	---

Sequor's Key Advantage — Cost Efficiency

With Sequor, you get **25X more Scenario Data at approximately 30% Lower Cost than Real-Data Competitors**

Same Task
“Pick and place”

Pricing Comparison by Data Collection Method

Real Data Collection average pricing **Competitor**
Teleoperation-based — human-operated, scenario-by-scenario

Category	Item	Unit	Price
Setup	Teleoperation hardware	Per initial setup	\$40,000+
Operation	Skilled operator cost	Per hour	\$35 ~ \$100/hr
Collection	Per-scenario cost	Per scenario	\$0.42 ~ \$0.70
Environment	Environment build (on-site setup)	Per setup	\$5,000 ~ \$50,000+

AY-ROBOTS **\$3,060**
U.S. Operator Hourly Rate
47.6hrs × €55/hr = €2,619

Sequor Robotics Pricing 
Simulation-based — unlimited scenario generation after setup

Category	Item	Unit	Price
Environment	World file	Per sqm	\$5 / sqm
Setup	Robot file (per type)	One-time	\$5,000
Setup	Sensor plugin (per type)	One-time	\$1,000
Training	Gym environment	Per task	\$100
Collection	Scenario run (simulation)	Per scenario	\$0.03 / scenario

Simulation — Sequor Robotics **\$2,114**
Simulation 50,000 + Real 300
Simulation: 50,000 × \$0.03 = \$1,500
Real data supplement: \$614

* Most simulation companies use subscription models, while Sequor Robotics offers a more cost-efficient usage-based model.

Sequor's Key Advantage — Cost Efficiency (Future Use Case Ver.)

Home Environment

Cooking Assistance

Estimated Cost for Real Data Collection

Category	Item	Quantity	Price
Environment	Environment Setup (On-site)	10	\$15,000/each
Collection	Data Collection	160 tasks	\$3,060/task
			\$639,600

Estimated Cost for Sequor Robotics

Category	Item	Quantity	Price
Environment	World File(50sqm)	100	\$5 / sqm
Initial Setup	Robot · Sensor Plugin · 3D Asset	(1 / 2 / 100 units)	\$8,000
Learning	Gym Env	160 tasks	\$100/task
Collection	Scenario (Simulation)	160 tasks	\$2,110/task
			\$386,600

Logistics DAS

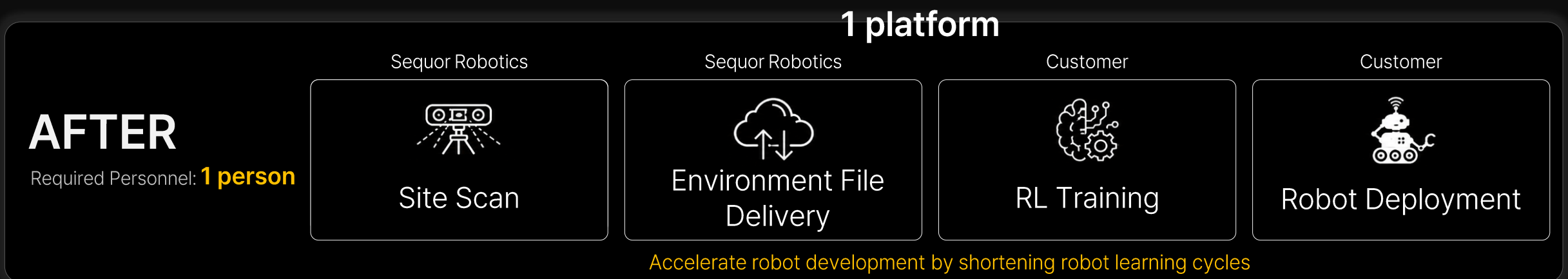
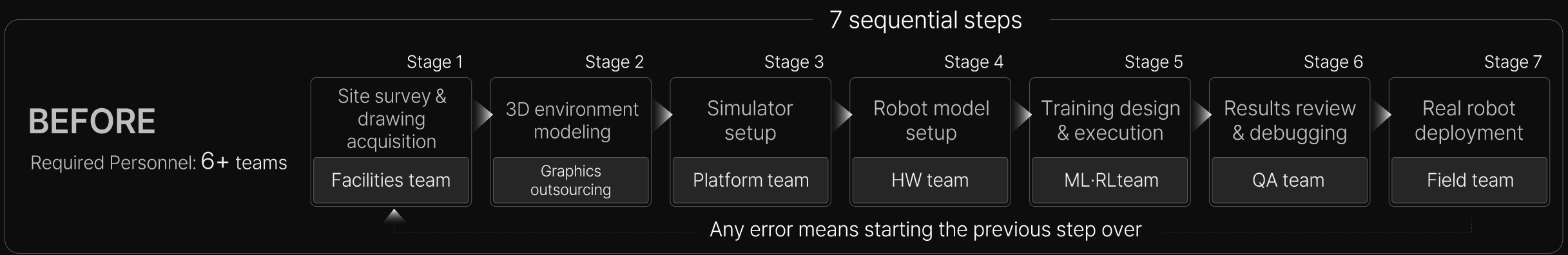
Order Picking & Packing

Category	Item	Quantity	Price
Environment	Environment Setup (On-site)	1	\$35,000/each
Collection	Data Collection	200 tasks	\$3,060/task
			\$647,000

Category	Item	Quantity	Price
Environment	World File(500sqm)	1	\$5 / sqm
Initial Setup	Robot · Sensor Plugin · 3D Asset	(1 / 2 / 135 units)	\$8,350
Learning	Gym Env	200 tasks	\$100/task
Collection	Scenario (Simulation)	200 tasks	\$2,110/task
			\$452,850

Sequor's Key Advantage — Time Savings

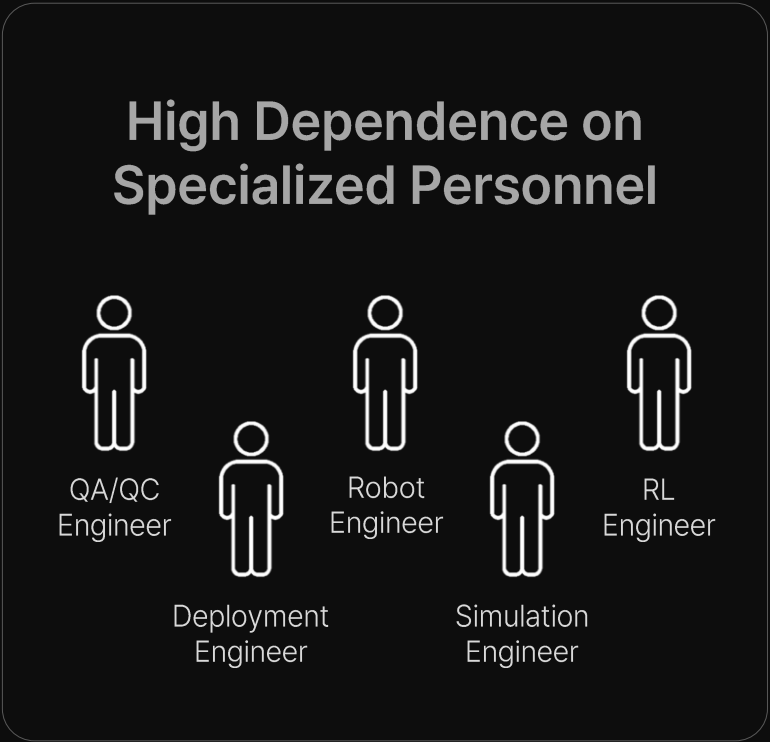
From multi-team handoffs → one person, one platform



Sequor's Key Advantage — Lower Barrier to Entry

Making Complex Robot Development Easier

Existing Approach



Sequor Robotics



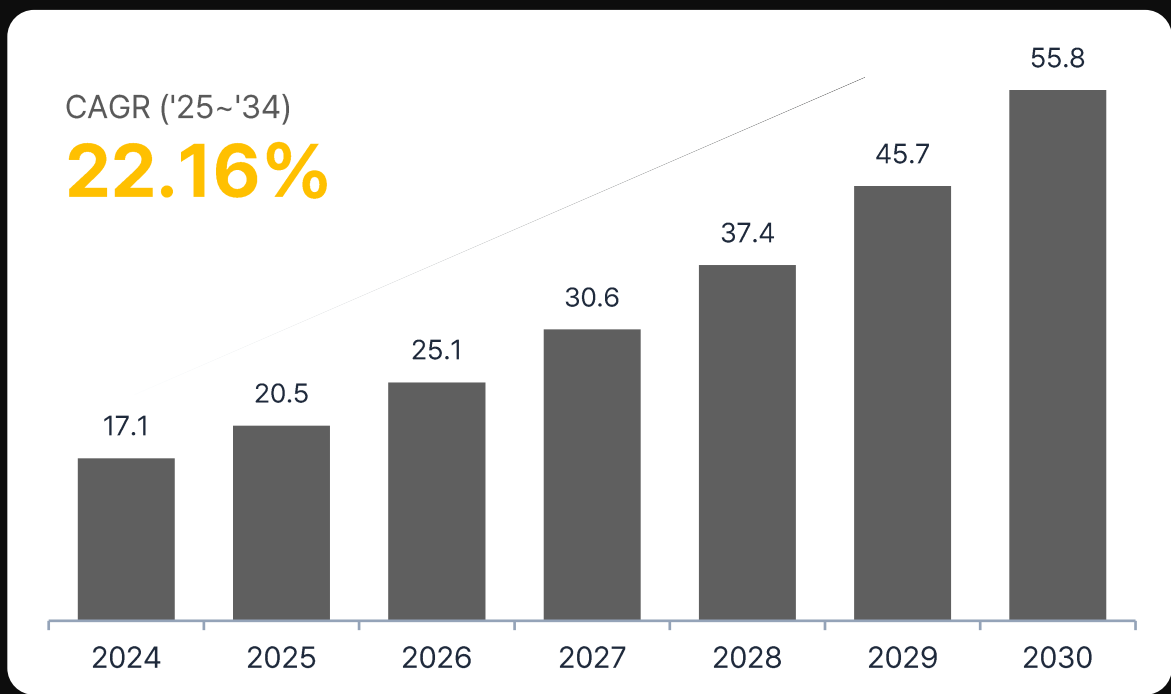
A platform that lowers the barrier to simulation adoption
by eliminating complex multi-team workflows.

Market Outlook

The AI robot training market is exploding — simulation is at the core.

AI in Robotics Market Size

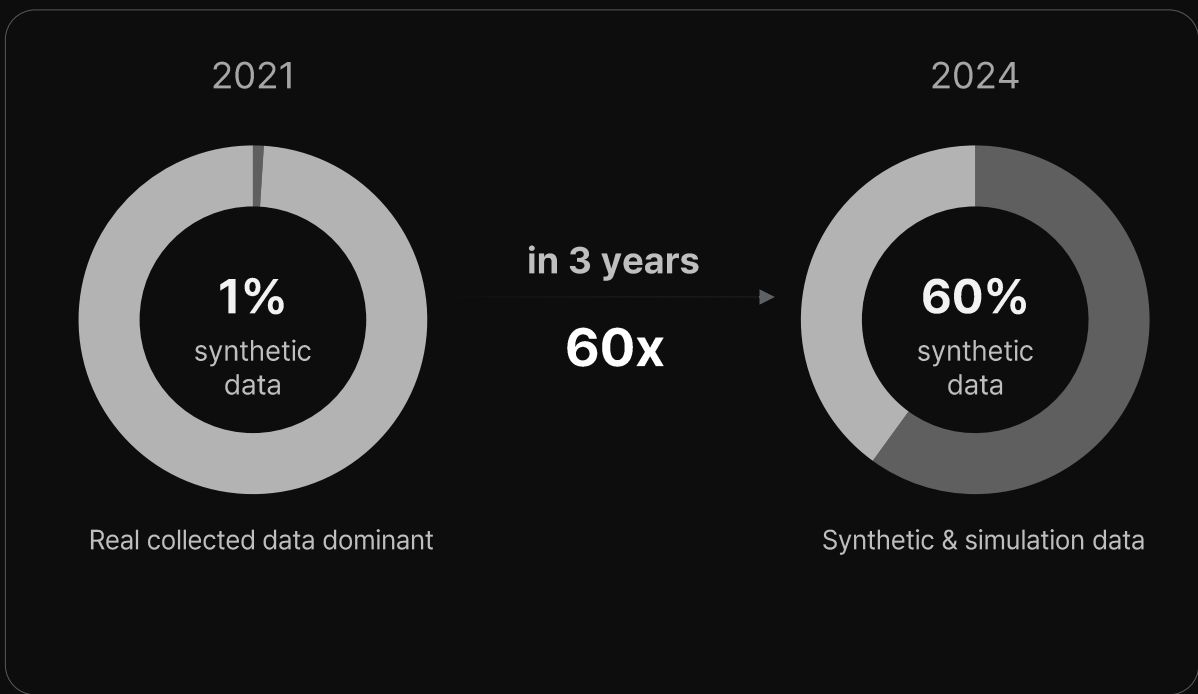
(Unit: USD Billion)



*Source: Precedence Research, 2025

A Paradigm Shift in AI Training Data

Synthetic & simulation-based data is becoming the dominant source of training data



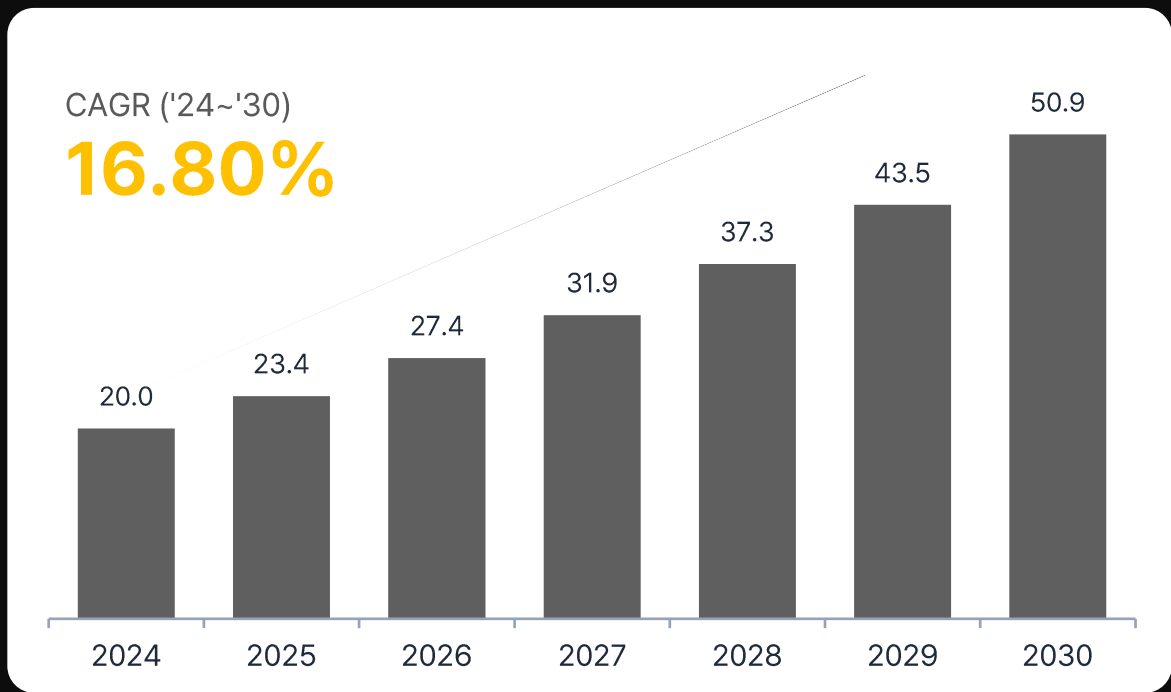
*Source: Gartner

Market Outlook

The robot simulation market is shifting toward software

Robot Simulation Market Size

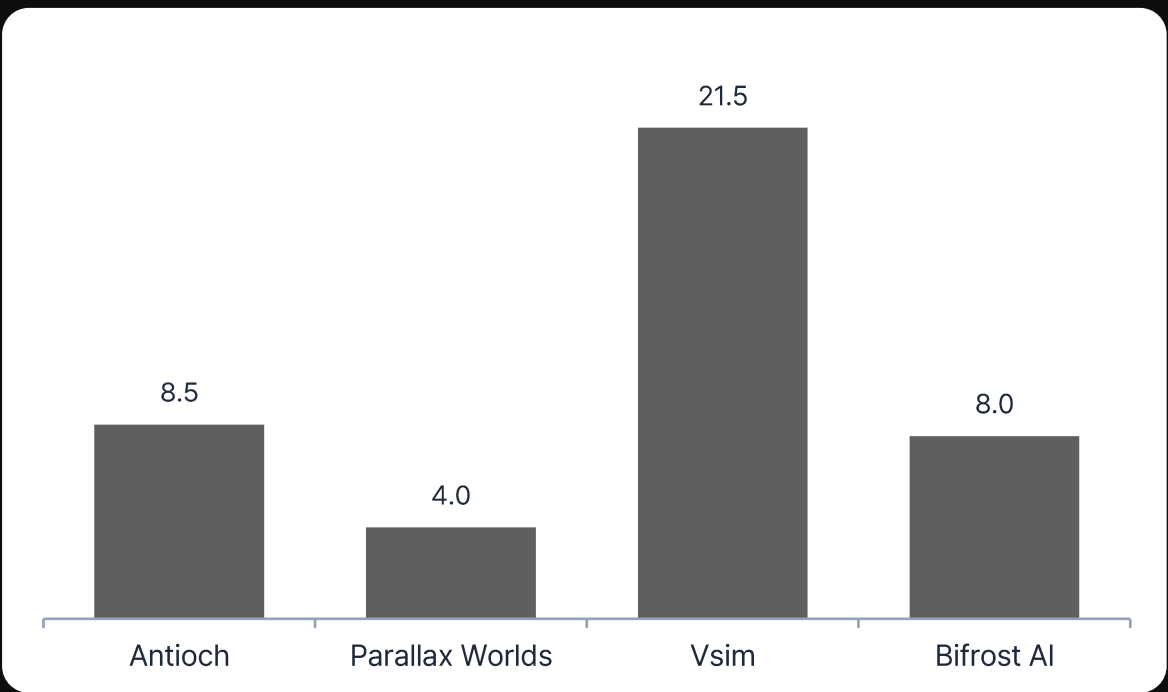
(Unit: USD Billion)



*Source: Cognitive Market Research, 2024

Investment Trends in the Robot Simulation Market

(Unit: USD Million)



Business Model

Revenue Structure

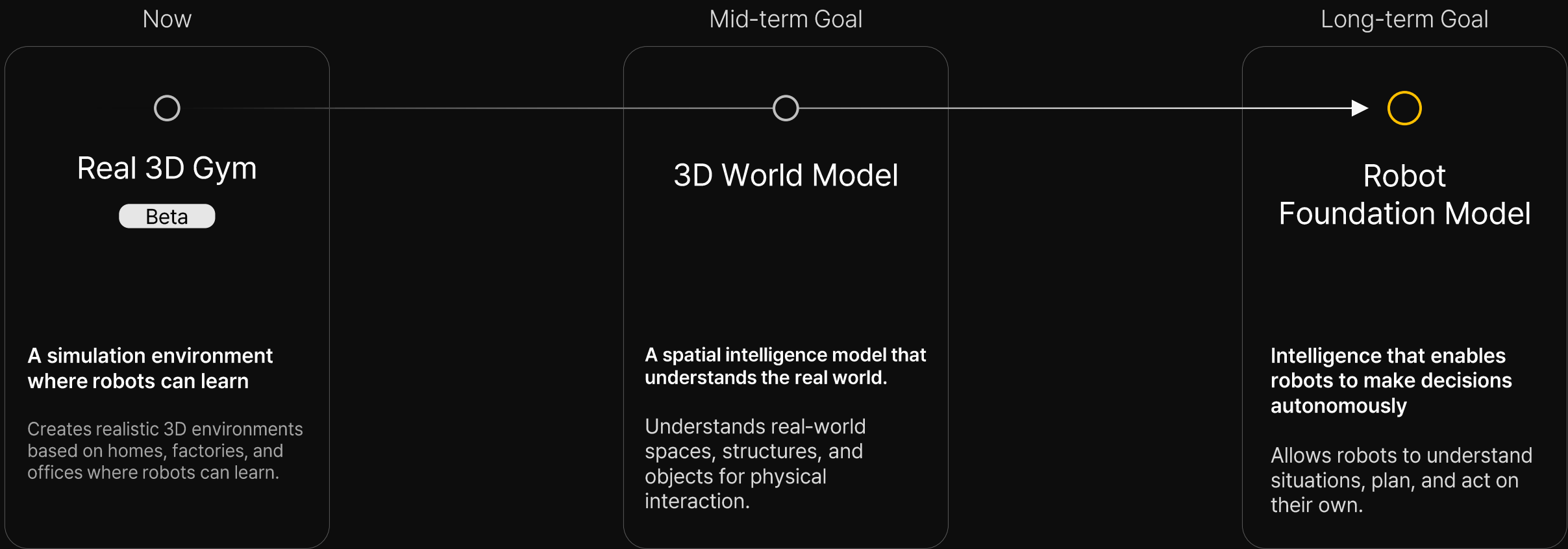


50,000 scenarios are required per task → \$1,500 to complete training for a single task

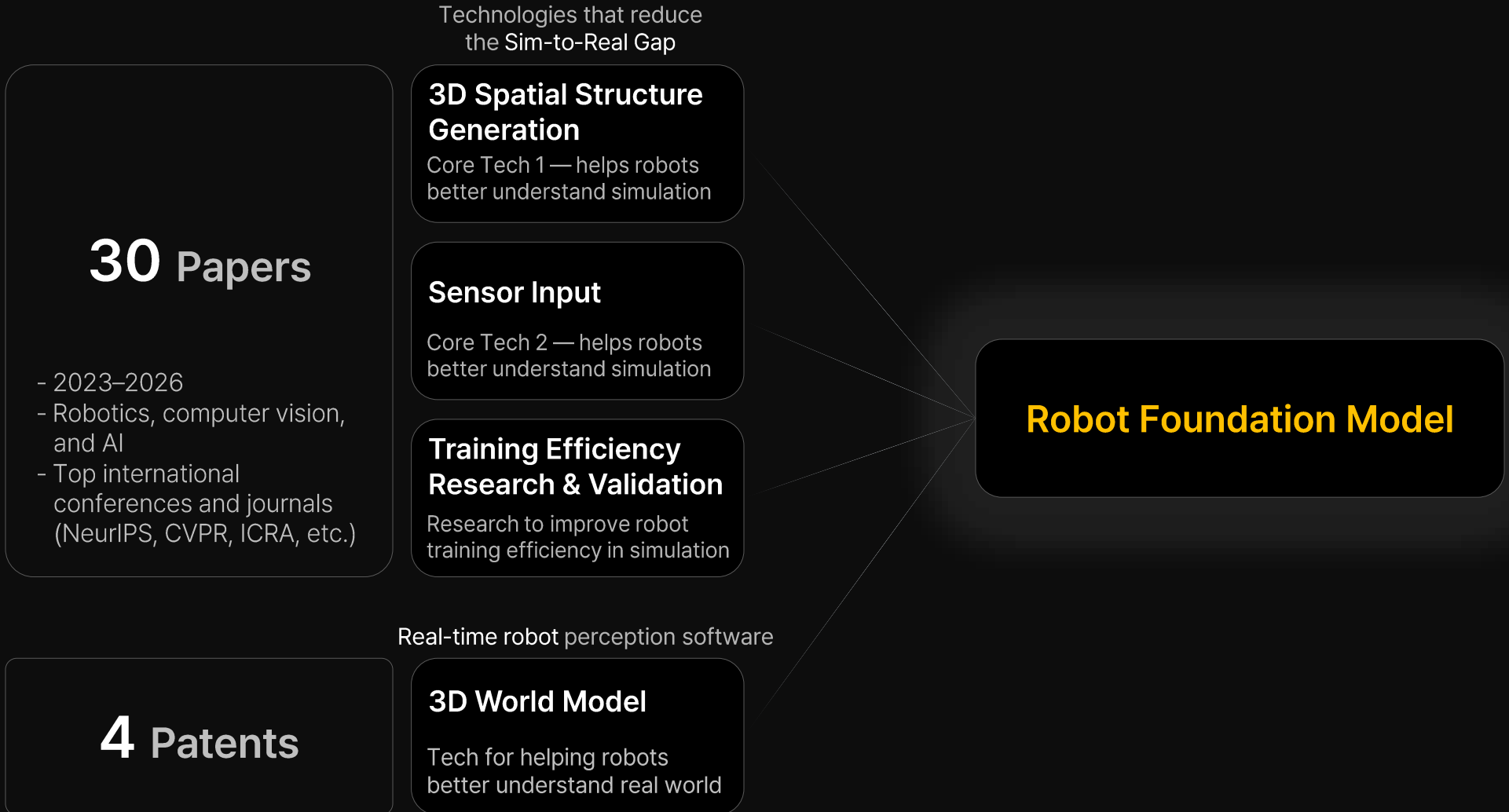
Product	Price	Description
World File	\$5 / sqm	Real-world environment scanning · Digital twin generation
Robot File	\$5,000 / robot type	Custom
Sensor Plugin	\$1,000 / sensor type	Custom
3D Asset	\$10 / asset	Object and environment asset generation
Gym Environment	\$100 / task	Task training environment design
Scenario Execution	\$0.03 / scenario	Learning scenario generation and execution

Vision

A world where robots act on their own with just a single word



Sequor's Technical Edge



Sequor's Technology 1

3D spatial structure optimized for robot learning

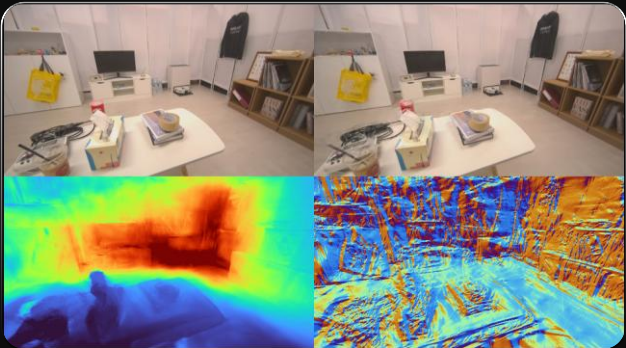
Robots don't just need visually appealing environments — **collision accuracy, distance, and spatial structure** must be precise for high-performance learning.

Comparison of learning suitability across commonly used environment tools

Method	Collision Geometry (robot collision & distance accuracy)	Visual Quality (real-world fidelity)	Robot learning Suitability
Visual SLAM (Matterport)	✔ Strong	✘ Weak	Partially suitable
Neural Rendering (NVIDIA)	✘ Weak	✔ Strong	Partially suitable
Sequor 3D Space Scan	✔ Strong	✔ Strong	✔ Suitable

Sequor has developed 3D spatial generation technology that satisfies both visual quality and physical structure for robot learning.

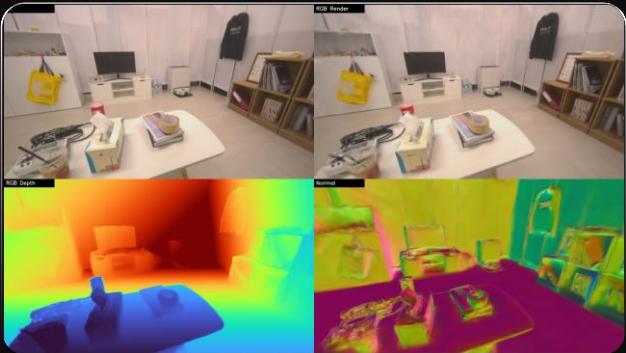
NVIDIA



Visual quality-focused environment reconstruction



Sequor Robotics



Accurate spatial structure reconstruction optimized for robot learning

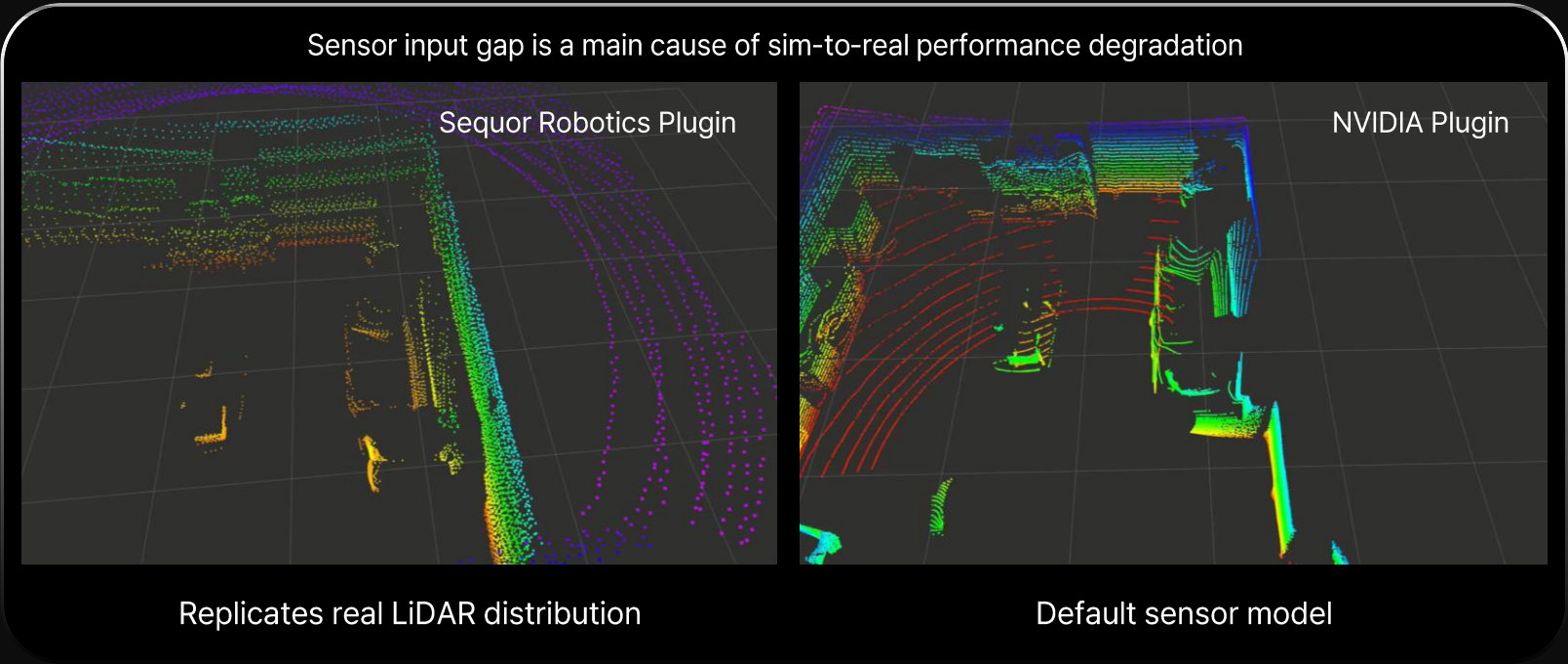
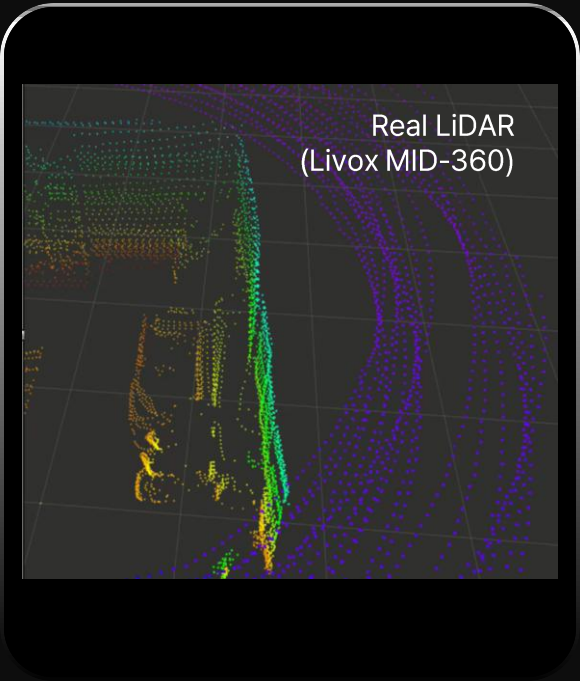


Accurate 3D structure directly impacts sim-to-real performance and learning stability

Sequor's Technology 2

Sensor data that replicates real-world conditions

Many robot engineers say the gap between simulation and real sensor data requires constant extra work.



Sequor has built input data that closely replicates real sensor characteristics to improve training performance and deployment stability.

Sequor's Technology 3

Simulation learning that transfers reliably to the real world

Sequor has directly validated through research that robot policies learned in simulation perform stably in real-world environments. Combining real and simulation data improves training performance (Published at IROS 2025)

Learning Data	Success Rate
Real	30%
Simulation	60%
Hybrid (Real + Simulation)	75%

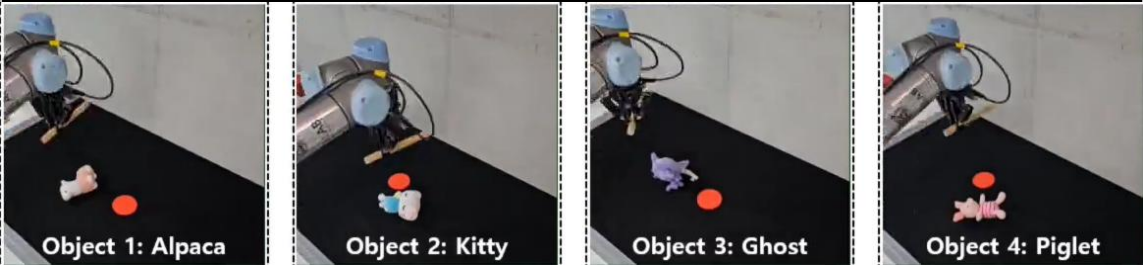
Navigation Task

Environment	Push to target	Wiping	Stick pushing
Simulation	96.3%	91.2%	86.9%
Real	92.5%	87.5%	85%
Gap	3.8%	3.7%	1.9%

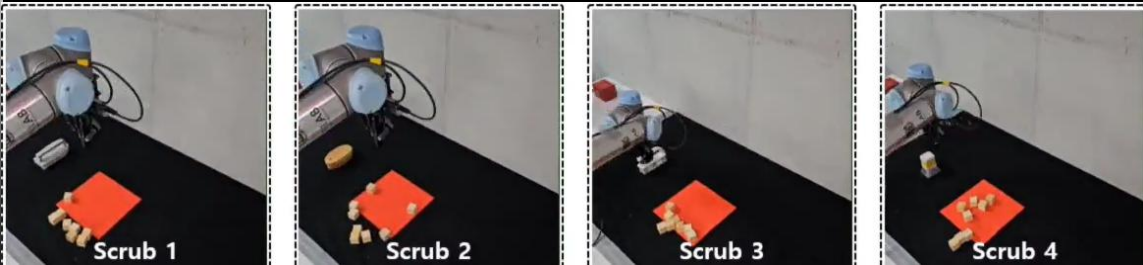
Simulation data alone achieves high success in real-world

Manipulation Task

Push to target



Wiping



Sequor's Technology 4

3D World Model

An AI model that recognizes space in real time using only a single camera and generates a 3D map — a core foundation for Robot Foundation Models.



Existing 3D AI struggled with real-time processing because spatial data is extremely large.

Sequor solves this by dramatically reducing computation, enabling real-time operation even on low-power embedded boards.

3D Mapping
Precisely reconstructs indoor spaces in 3D using only a single camera — no LiDAR required.

Visual Localization
Instant localization from a single image

98%+
Localization accuracy

Real-Time Acceleration
1.5 FPS → 4.7 FPS achieved
Targeting 10 FPS for commercialization

Sequor's Technology 4

3D World Model

Performance Comparison

Category	Existing Method (LiDAR + Depth)	Sequor 3D World Model
Required sensors	Multiple sensors	Single RGB camera
Spatial resolution	5cm	Up to 0.8mm
Recovery time after location drift	Remapping required	1~5 seconds
Product cost	High multi-sensor cost	Up to 15% cost reduction

25.11.18. Sequor technology demo (Samsung, Namuga, Pixelplus visited)

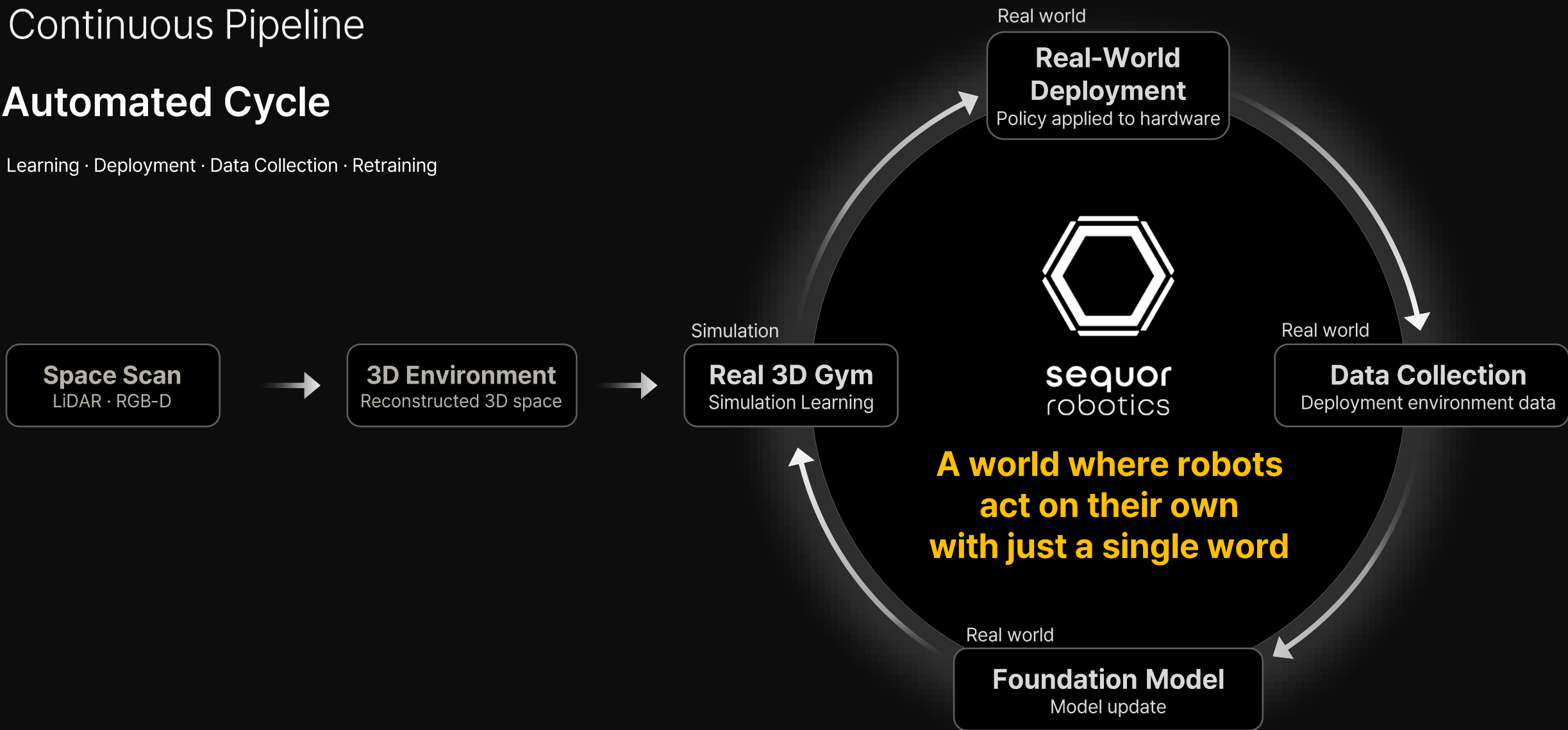
	Secured \$2.1M in R&D funding over 2 years Established a four-party consortium
	Signed a service agreement Completed visual localization accuracy validation

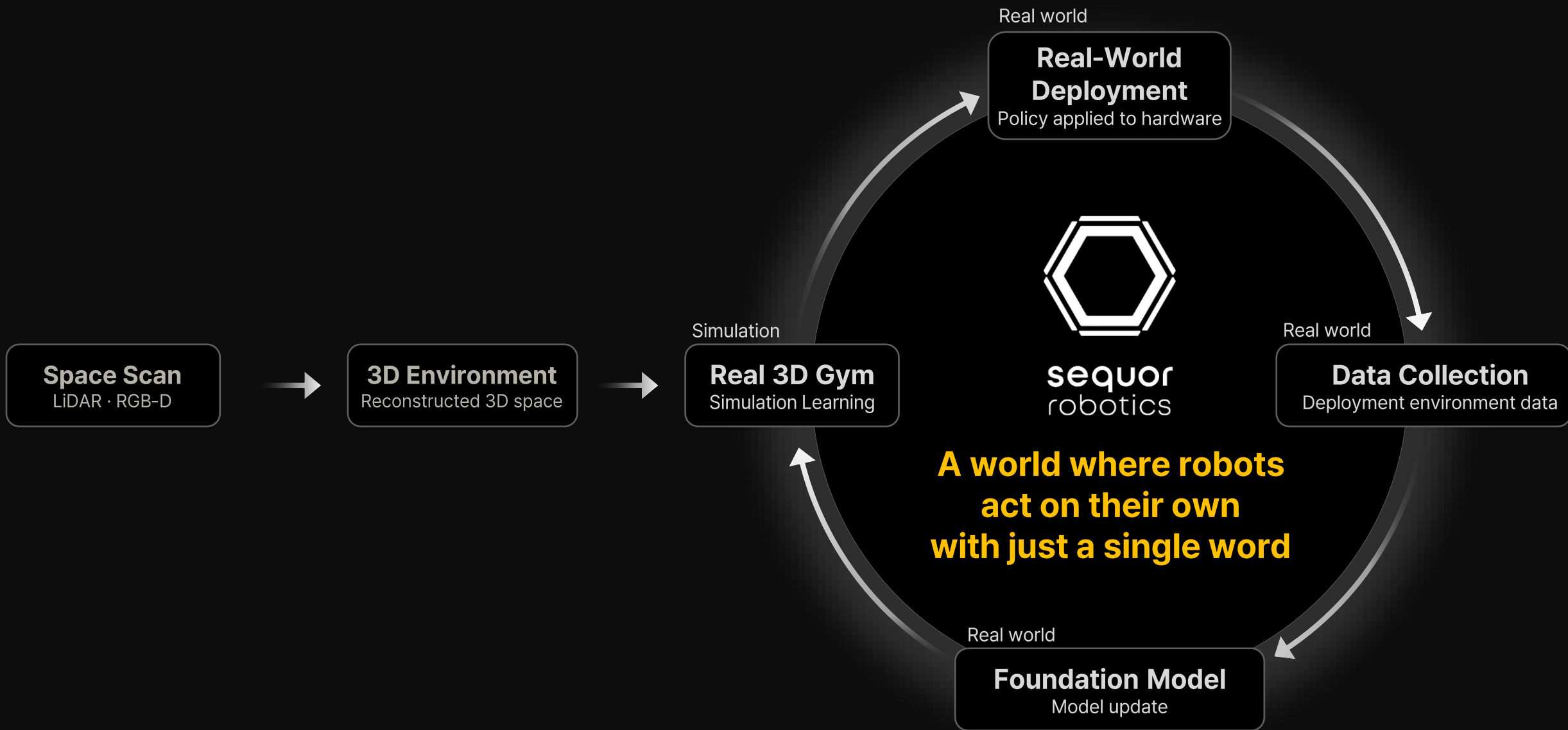


Continuous Pipeline

Automated Cycle

Learning · Deployment · Data Collection · Retraining





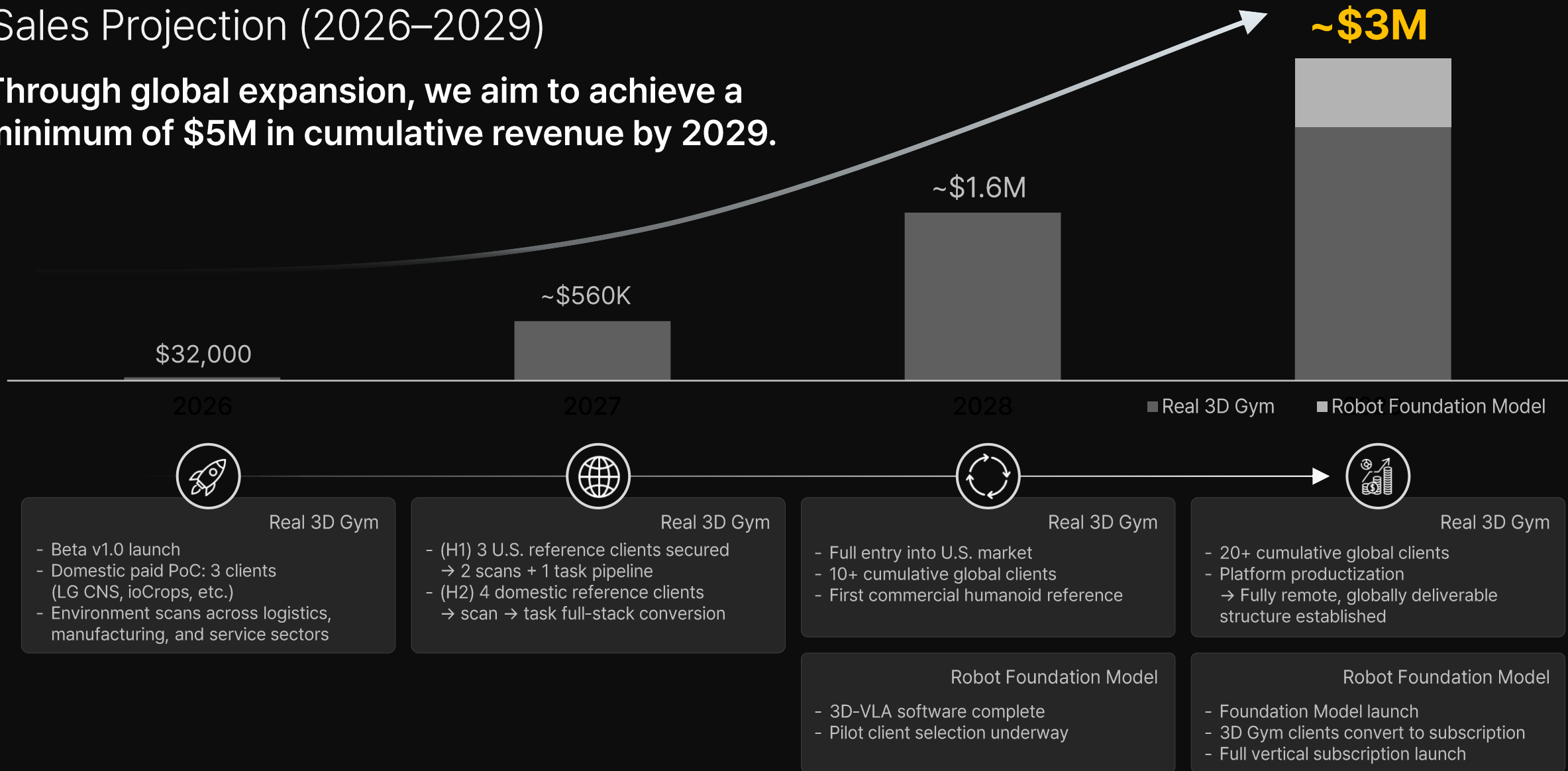
Milestones

- | | |
|---------|--|
| 2026 H2 | Real 3D Gym beta launch & domestic PoC secured
U.S. exhibitions and accelerators → U.S. customer pipeline established <ul style="list-style-type: none">- IBK Changong- Physical AI Superstart Begin partnerships with U.S. companies
(In partnership discussions with a real data company) |
| 2027 H1 | Real 3D Gym product launch <ul style="list-style-type: none">- Commercialization of high-demand technologies (Automated environment setup & scenario generation) PoC to Paid Conversion
Securing Domestic Clients |
| 2027 H2 | U.S. entity incorporated & initial local team in place
U.S. paying customers secured |
| 2028 H1 | 3D-VLA software product launch
Series B fundraise |



Sales Projection (2026–2029)

Through global expansion, we aim to achieve a minimum of \$5M in cumulative revenue by 2029.



TEAM



Head of Business, CEO & Co-Founder

Jeongwoo Oh

Ph.D. in Robot Learning, Seoul National Univ.

B.S. in Electrical and Computer Engineering,

Seoul National Univ.

Forbes 30 Under 30, Korea/Asia



Head of Technology, CTO & Co-Founder

Songhwai Oh

Professor, Robot Learning Lab,

Seoul National Univ.

Ph.D. in Electrical Engineering &

Computer Science, UC Berkeley

Former Senior Engineer, Intel & Synopsys



Building products and conducting research
with a team of specialists across multiple fields

Product Lead

Expertise: Robot Application

#SNU M.S. #CLOBOT #Siemens

AI/Robotics Researcher

Expertise: Latent 3D World Model, Neural 3D Map

#SNU Ph.D.

AI/Robotics Researcher

Expertise: Real-to-Sim-to-Real

#SNU Ph.D.

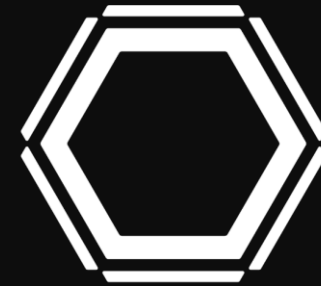
3D Vision Researcher

Expertise: 3D reconstruction, Vision AI

#Camera Hardware Specialist #KU B.S.

Company History

- 2022** Dec: Sequor Robotics incorporated
- 2023** Mar: Seed investment from Future Play
Jul: Selected as a Deeptech Tips company
- 2024** Mar: Selected for DIPS 1000+
Jul: Selected for Startup Autobahn Korea (LG Electronics partnership program)
Aug: Seed investment from BA Partners
Nov: Selected for LG SuperStartup Cohort 3
Nov: NDA signed with Samsung Electronics
- 2025** Apr: NDA signed with Samsung Electronics America
May: Patent filed — 3D spatial semantic occupancy prediction
Jun: Selected for IBK Changong Accelerating Program
Jul: PoC contract signed with LG Electronics
Aug: MOU signed with Samsung Electronics; registered as Samsung Electronics supplier
Aug: Patent filed — cleaning path optimization
Sep: Patent filed — 3D map management
Nov: Participated in Samsung Electronics AI Alliance
Dec: Patent filed — bird's-eye-view-based robot path planning
(Samsung Electronics, Namuga, Pixelplus, Sequor Robotics)
Nov: Selected for Deeptech Value Program (LG Electronics HS Division)
- 2026** Jan: Patent registered — 3D spatial semantic occupancy prediction
May: Patent registered — sparse octree-based semantic voxel inference



sequor
robotics



Scaling Physical AI through Robot Foundation Models

2026 Investor Relations